Srikriti Mutyala

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Education

University of Illinois Urbana-Champaign — Urbana, Illinois
Bachelor of Science in Computer Science and Statistics — 2028
GPA: 3.93
Relevant coursework: Data Structures, Discrete Structures, Linear Algebra, Statistical Analysis, Calculus III

Experience

Undergraduate Student Researcher — Urbana, Illinois
Professor Rabbi's Lab at UIUC — February 2026 - Present
• Developed ML workflows to transform Fitbit sleep records into behavioural embeddings to discover sleep patterns across thousands of patients
• Implemented an AI-driven sleep intervention framework leveraging 18 biomarkers and Fitbit datasets from 54,000+ participants.
• Contributed to the development of a personalized sleep recommendation system by analyzing 100+ participant sleep profiles and identifying behavior patterns associated with sleep deficiency and irregularity.

Undergraduate Student Researcher — Urbana, Illinois
XAbility Lab — January 2026 - Present
• Deployed an AI-assisted accessibility platform for low-vision learners, integrating React, Node.js, Flask, YOLOv8, OpenCV, and Mistral AI to support exploration of tactile statistical plots.
• Engineered a computer vision pipeline that extracts 5+ statistical points from boxplots, generating representations for guidance.
• Built a 3-service architecture for real-time image analysis, hand tracking, and conversational AI across frontend, backend, and perception layers.

Project Manager & Full-Stack Developer — Urbana, Illinois
NOBE Illinois — January 2026 - Present
• Led a team of 6 software engineers over a 2-semester development cycle, delivering a platform serving 120+ members and 10+ executive board members while coordinating sprint planning, GitHub issue management, and feature delivery.
• Developed a full-stack attendance and engagement platform using Next.js, TypeScript, and Supabase, automating event tracking, member management, Google Calendar integration, and QR-based check-in workflows.
• Engineered automated attendance auditing and strike enforcement systems using Docker, Vercel Cron Jobs, and Resend SMTP, streamlining organizational operations and policy compliance.

Student intern — Chicago, Illinois
Discover Partners Institute — June 2025- August 2025
• Analyzed 9 urban datasets using Python, Pandas, and scikit-learn to investigate emissions trends impacting the Chicago metropolitan area.
• Built predictive machine learning pipelines utilizing Random Forest and Neural Network models to identify environmental risk factors and evaluate potential intervention strategies.
• Communicated findings through presentations, culminating in a final showcase at the Google Chicago office for community stakeholders.

University Projects

GitHub Projects Self-Writer — June 2026- Present
• Built an end-to-end AI coding agent that automatically detects when a GitHub Project ticket is moved to "Ready", uses the Gemini AI API to generate production code based on the ticket description, and opens a Pull Request — all without human intervention
• Architected a serverless event-driven pipeline using Cloudflare Workers as webhook middleware, GitHub Actions for CI/CD orchestration, and the GitHub GraphQL API for project data retrieval

Kuchipudi Hand Gesture Teaching Tool — January-November 2025
•Developed a CNN-based gesture recognition system for 31 Kuchipudi hand gestures, training on 3,100+ self-curated images and achieving 93% accuracy.
• Built a real-time computer vision pipeline using MediaPipe, OpenCV, and Flask, leveraging hand-landmark extraction for robust classification
• Deployed an educational web application used by 100+ dance students, providing instant gesture recognition and instructional feedback.

Stock prediction app, Girl Who Code financial app challenge, finalist
• Developed an LSTM-based stock prediction platform using 15+ years of market data, achieving 94% forecasting accuracy.
• Built machine learning pipelines with TensorFlow, Keras, and NumPy to generate stock forecasts, buy signals, and market visualizations
• Designed and deployed a Streamlit dashboard for financial analysis, becoming a Girls Who Code Financial Challenge Finalist.

Skills

- **Languages**: Python, Java, JavaScript, TypeScript, SQL
- **Frameworks & Tools**: React, Next.js, Flask, TensorFlow, Scikit-learn, Pandas, OpenCV, Docker, Git, GitHub Actions
- **Areas**: Machine Learning, Deep Learning, Computer Vision, Full-Stack Development, Data Analysis,